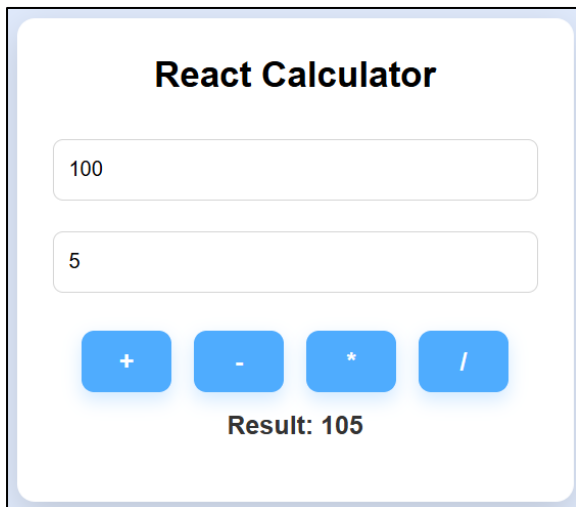


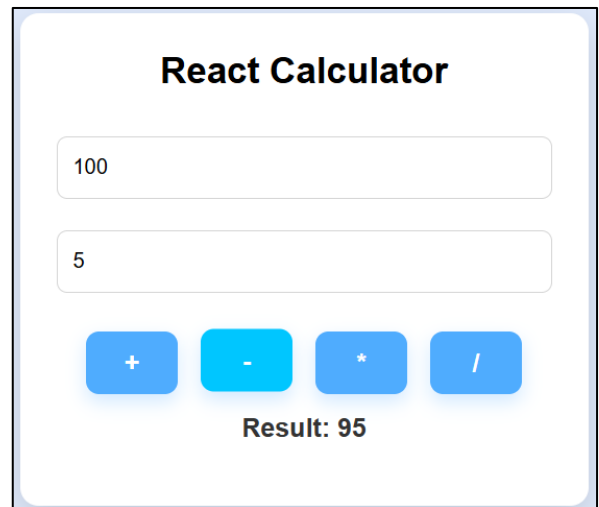
Practical No. 4

Aim: To design and develop a calculator application using React JS that performs basic arithmetic operations using functional components and state management.

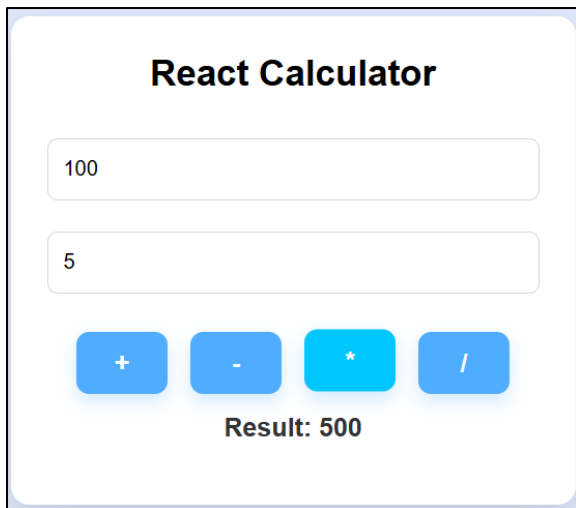
Practical No. 4 Output:



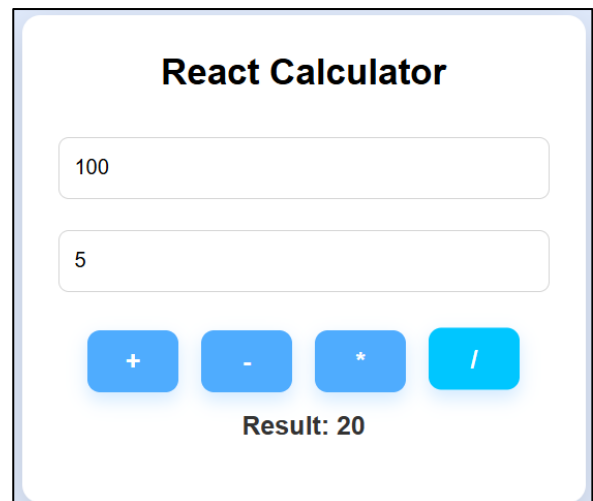
The screenshot shows a web application titled "React Calculator". It has two input fields: the first contains "100" and the second contains "5". Below the inputs are four buttons: "+", "-", "*", and "/". The "+" button is highlighted in a darker blue. Below the buttons, the text "Result: 105" is displayed.



The screenshot shows the same "React Calculator" app. The "-" button is now highlighted in a darker blue. The result displayed below the buttons is "Result: 95".



The screenshot shows the same "React Calculator" app. The "*" button is now highlighted in a darker blue. The result displayed below the buttons is "Result: 500".



The screenshot shows the same "React Calculator" app. The "/" button is now highlighted in a darker blue. The result displayed below the buttons is "Result: 20".

Practical No. 4 Code:

App.js:

```
import React, { useState } from "react";
import "./App.css";
function App() {
  const [num1, setNum1] = useState("");
  const [num2, setNum2] = useState("");
  const [result, setResult] = useState("");
  const calculate = (operation) => {
    const a = parseFloat(num1);
    const b = parseFloat(num2);
    if (isNaN(a) || isNaN(b)) {
      setResult("Invalid Input");
      return;
    }
    switch (operation) {
      case "+":
        setResult(a + b);
```

```
        break;
      case "-":
        setResult(a - b);
        break;
      case "*":
        setResult(a * b);
        break;
      case "/":
        setResult(b !== 0 ? a / b : "Cannot divide by
zero");
        break;
      default:
        break;
    }
  };
  return (
    <div className="calculator">
      <h2>React Calculator</h2>
```

```

<input
  type="number"
  placeholder="Enter first number"
  value={num1}
  onChange={(e) => setNum1(e.target.value)}
/>
<input
  type="number"
  placeholder="Enter second number"
  value={num2}
  onChange={(e) => setNum2(e.target.value)}
/>
<div className="buttons">
  <button onClick={() =>
calculate("+")}>+</button>
  <button onClick={() => calculate("-")}>-
</button>
  <button onClick={() =>
calculate("*")}>*</button>
  <button onClick={() =>
calculate("/")}>/</button>
</div>
<h3>Result: {result}</h3>
</div>
);
}
export default App;

```

App.css:

```

body {
  background: #e0eafc;
  font-family: Arial, Helvetica, sans-serif;
  margin: 0;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
}
.calculator {
  width: 380px;
  padding: 28px;
  background: white;
  border-radius: 14px;
  box-shadow: 0 10px 30px rgba(0,0,0,0.15);
  text-align: center;
}
.calculator h2 {
  margin: 0 0 22px 0;
  font-size: 28px;
  font-weight: 700;
}
.calculator input {

```

```

width: 100%;
padding: 14px 12px;
margin: 12px 0;
font-size: 16px;
border-radius: 8px;
border: 1px solid #d6d6d6;
outline: none;
box-sizing: border-box;
transition: 0.2s ease;
}
.calculator input:focus {
  border-color: #4facfe;
  box-shadow: 0 0 0 3px rgba(79, 172, 254, 0.25);
}
.buttons {
  display: flex;
  justify-content: center;
  gap: 18px;
  margin: 18px 0 10px;
}
.buttons button {
  width: 70px;
  height: 48px;
  font-size: 20px;
  font-weight: 600;
  background: #4facfe;
  color: white;
  border: none;
  border-radius: 10px;
  cursor: pointer;
  transition: 0.2s ease;
  box-shadow: 0 6px 14px rgba(79, 172, 254, 0.25);
}
.buttons button:hover {
  background: #00c6ff;
  transform: translateY(-2px);
}
.buttons button:active {
  transform: translateY(0px);
}
h3 {
  margin-top: 14px;
  font-size: 20px;
  color: #333;
}
h3 span {
  color: #4facfe;
  font-weight: 700;
}

```